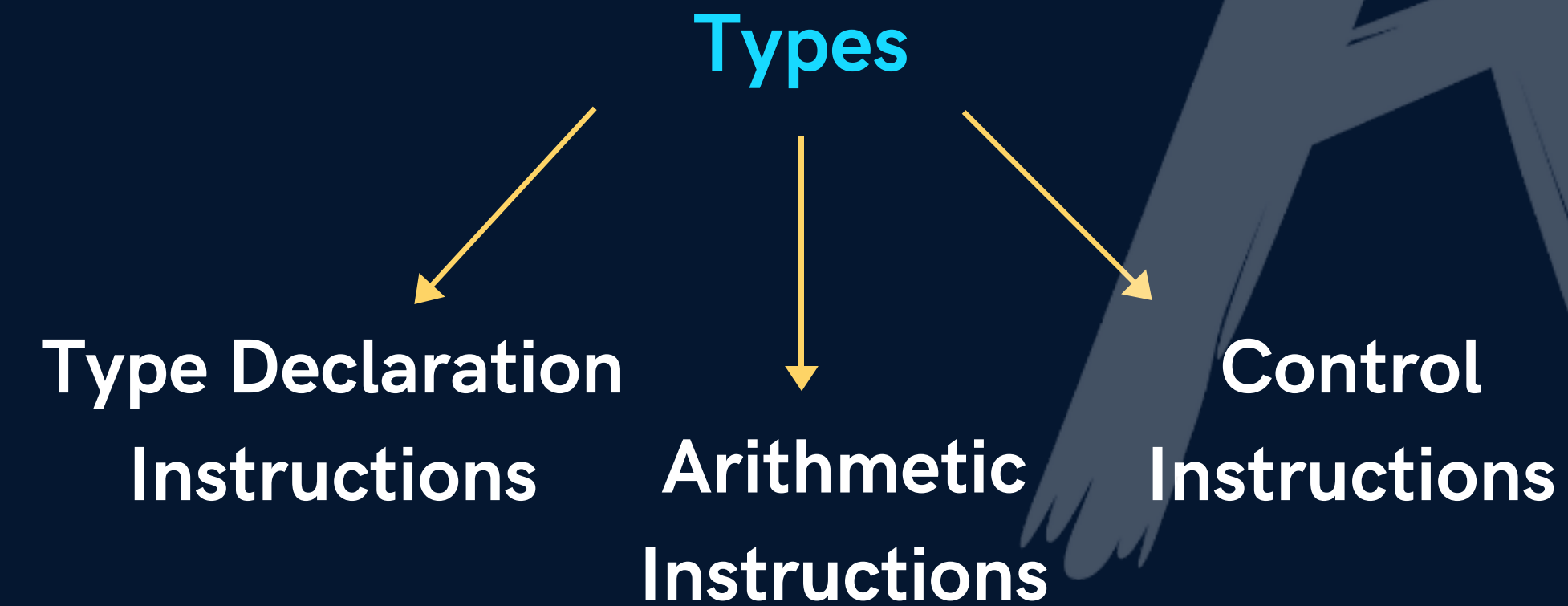


Instructions

These are statements in a Program



Instructions

Type Declaration Instructions → Declare var before using it

VALID

```
int a = 22;  
int b = a;  
int c = b + 1;  
int d = 1, e;
```

```
int a,b,c;  
a = b = c = 1;
```

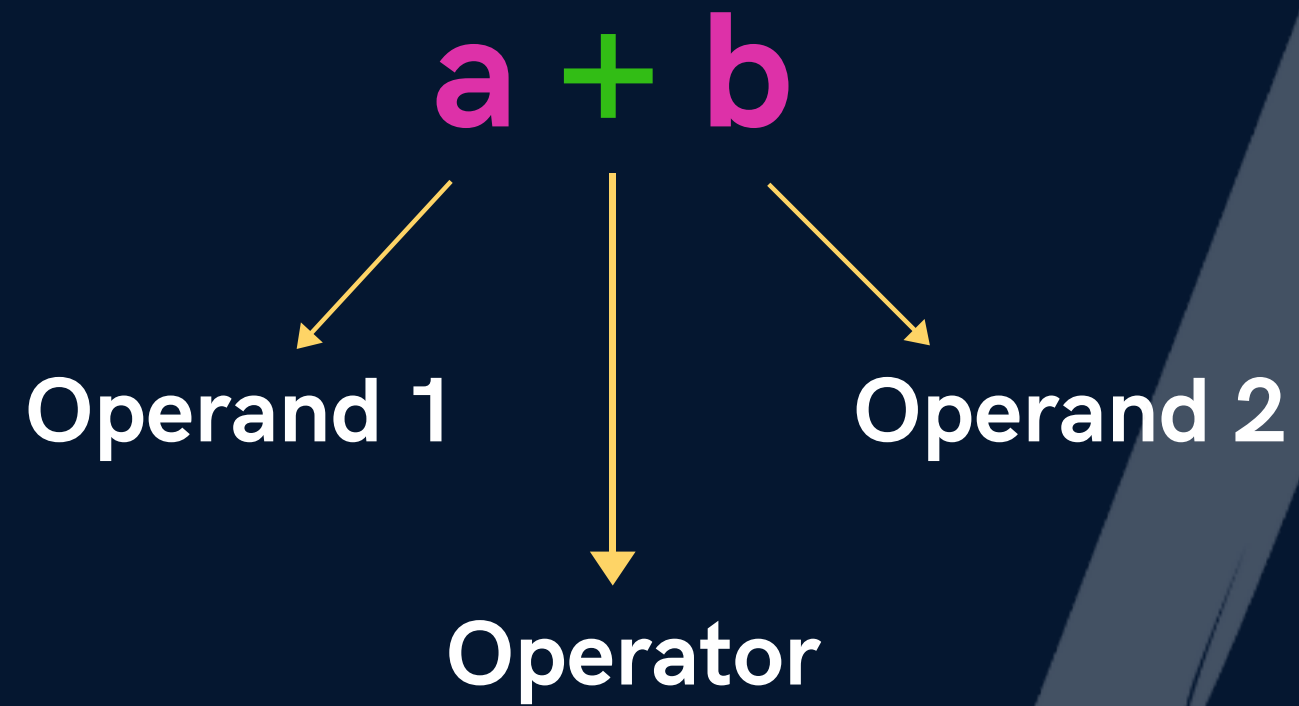
INVALID

```
int a = 22;  
int b = a;  
int c = b + 2;  
int d = 2, e;
```

```
int a,b,c = 1;
```

in valid section all
valid instructions
are given while
invalid
instructions are
given in the other
section

Arithmetic Instructions



NOTE - single variable on the LHS

Arithmetic Instructions

VALID

$a = b + c$

$a = b * c$

$a = b / c$

INVALID

$b + c = a$

$a = bc$

$a = b^c$

$\wedge$ this is bitwise operator it does not gives power

NOTE - $\text{pow}(x,y)$ for x to the power y

Arithmetic Instructions

★ Modular Operator %

Returns remainder for int

$$3 \% 2 = 1$$

$$-3 \% 2 = -1$$

Arithmetic Instructions

Type Conversion

int op int → int

int op float → float

float op float → float

int can be stored in float
but float can't be stored
in int as int takes 2 bytes
and float takes 4 bytes
float is larger

Arithmetic Instructions

Operator Precedence

$*, /, \%$



$+, -$



$=$

$$x = 4 + 9 * 10$$

$$x = 4 + 90$$

$$x = 94 \text{ (correct)}$$

$$x = 130 \text{ (incorrect)}$$

$$x = 4 * 3 / 6 * 2$$

like the bodmas here we will solve higher will be the precedence thus we will solve

Arithmetic Instructions

Associativity (for same precedence)

Left to Right

$$x = 4 * 3 / 6 * 2$$

$x=4$ (correct)

When same precedence is there then associaty rule comes into play

for ex +,-
*,/,%

Instructions

Control Instructions

Used to determine flow of program

a. Sequence Control

b. Decision Control

c. Loop Control

d. Case Control

flow la alter
karayachi power
thevtat

Operators

a. Arithmetic Operators

$+, -, /, *, \%$

b. Relational Operators

c. Logical Operators

d. Bitwise Operators

e. Assignment Operators

f. Ternary Operator



Operators

Relational Operators

==

>, >=

<, <=

!=

if 1 is printed means it is true and if 0 is printed means it is false. (2,4,5,6,7,-1 and all other integer means true in C.

Operators

Logical Operators

&& AND

|| OR

! NOT

single & is bitwise
operator and && is
logical operator



Operator Precedence

operator precedence
tharnar logical
operator kashe
execute karayache te

Priority

Operator

1

!

2

*, /, %

3

+, -

4

<, <=, >, >=

5

==, !=

6

&&

7

||

8

=

Not
Arithemati
Relational
Logical

Operators

Assignment Operators

=

+=

--

*=

/=

%=

